Medford
Technologies

User Manual

Bluvia Neo

Washer-Disinfector

Contents

1. General Information

Please read this user manual carefully before commissioning the device.
The manual includes important safety instructions.For safe operation, please read this user manual thoroughly before commissioning the device. This manual contains crucial safety instructions.

1.1 Symbols Used

Symbol	Explanation
	Warning: Risk of personal injury if ignored
	Information: Important for proper operation
	Service – Maintenance task, qualified personnel only

1.2 Disposal

Our medical devices are built for quality and longevity. When it's time to decommission your unit, please dispose of it through authorized medical equipment recyclers. This ensures compliance with all relevant disposal specifications, especially for any potentially contaminated waste.

The packaging materials are chosen for their eco-friendly disposability and can be recycled according to local regulations. Recycling packaging reduces waste and conserves raw materials.

2. Safety Instructions

When operating the Bluvia Neo washer-disinfector, comply with the following safety instructions as well as those contained in subsequent chapters of this manual. The device may only be used for the intended purpose of cleaning and disinfecting medical instruments. Failure to comply with these safety instructions may result in personal injury, contamination risks, or damage to the device.

2.1 General Safety

- Operation of the washer-disinfector is permitted only by trained and competent personnel.
- The operator is responsible for ensuring that all users are regularly trained in the safe handling, loading, and operation of the device.
- Always wear protective gloves, laboratory coats, and approved safety eyewear when handling contaminated instruments, process agents, or waste.
- Do not bypass or disable safety features, including the automatic door interlocks.
- The device must never be used for purposes other than those described in this manual.

2.2 Installation Safety

- Inspect the device immediately after unpacking to ensure it has not suffered transport damage.
- Installation, commissioning, and servicing must only be carried out by authorized service technicians.
- Water supply and drainage connections must comply as per **ISO standards** requirements.
- Install the washer-disinfector in a dry, well-ventilated, and frost-free environment.
- The disconnection device (mains switch or plug) must remain easily accessible after installation.
- The device must not be operated in explosive atmospheres.
- Ensure that computer systems, USB sticks, or other documentation media connected to the device are protected from contact with liquids.

2.3 Process Agents

- Use only detergents, neutralizers, rinse aids, and regenerating salt that are approved by the manufacturer or specified in this manual.
- Process agents may contain hazardous or caustic substances. Always follow the manufacturer's safety data sheets (SDS) and wear appropriate protective equipment when handling chemicals.
- Store process agents only in their original containers, tightly sealed, and in a cool, dry place away from direct sunlight, heat sources, and moisture.
- Any liquid leakage inside the device may contain aggressive chemical residues. Handle spills with caution and follow standard safety protocols.

2.4 Maintenance Safety

- Always disconnect the washer-disinfector from the mains power supply before carrying out maintenance or service work.
- Routine maintenance, calibration, and repair may only be performed by authorized service technicians.
- Replace HEPA filters, gaskets, and seals only with manufacturer-approved spare parts to maintain validated performance and safety standards.
- Do not attempt to open or modify the device housing yourself. Unauthorized access to internal components can compromise warranty,electrical safety and performance validation.
- If repeated error messages occur, discontinue operation and contact authorized service support.

3. Intended Use & Performance

The **Bluvia Neo Washer-Disinfector** is a medical device intended for the **automated cleaning, disinfection, and drying of reusable medical instruments**. It is designed to support infection-control practices in hospitals, surgical centers, and dental clinics by providing standardized and validated reprocessing of critical and semi-critical instruments.

3.1 Intended Use

- **Instrument Types:**
  - General surgical instruments (solid, hinged, and disassemblable types)
  - Hollow-body instruments such as cannulas, suction tubes, and minimally invasive surgical instruments
  - Dental instruments (mirrors, probes, scalers, handpieces with adaptors)
  - Ophthalmic instruments requiring delicate but thorough cleaning
  - Stainless steel trays, DIN mesh baskets, and other reusable medical accessories
- **Process Scope:**
  - Pre-cleaning and removal of coarse soiling
  - Enzymatic cleaning and alkaline wash for effective removal of proteinaceous and organic residues
  - Neutralization of alkaline residues using low-pH neutralizers
  - High-temperature thermal disinfection to achieve validated microbial reduction
  - Forced-air drying with HEPA filtration to prevent recontamination and ensure immediate usability

3.2 Standards & Compliance

- Confirms to **ISO 15883-1** (General requirements for washer-disinfectors) and **ISO 15883-2** (Washer-disinfectors for surgical instruments).
- Manufactured under a certified **ISO 13485 Quality Management System**.
- Designed in alignment with hospital infection-control guidelines and international best practices.

3.3 Performance Characteristics

- **Thermal Disinfection:** Achieves validated disinfection with an **A0 value ≥ 3000**, which is adjustable based on selected program. This ensures effective inactivation of bacteria, viruses, and fungi, including resistant strains.
- **Drying System:** Integrated **HEPA H14-filtered forced-air drying** guarantees that instruments are delivered completely dry, reducing the risk of microbial regrowth and corrosion.
- **Water Quality Management: water softener** with regeneration cycle and **DI (deionized) water rinse** ensures stain-free, corrosion-free, and residue-free results.
- **Automated Process Monitoring:** Continuous monitoring of key parameters such as temperature, time, water conductivity, and detergent dosing guarantees repeatable and validated results for every cycle.
- **User Safety:** Automatic door lock prevents premature access during operation; drying air is filtered to prevent contamination of processed instruments.

3.4 Clinical Benefits

- Ensures **standardized reprocessing**, reducing human error compared to manual cleaning.
- Supports **traceability and documentation** via cycle logs and batch reports, aiding hospital audits and compliance.
- Minimizes staff exposure to contaminated instruments.
- Enhances longevity of medical instruments by preventing corrosion and mechanical wear from improper cleaning.

4. Device Description

4.1 External Overview

The Bluvia Neo features a **robust stainless steel construction** that ensures durability and resistance to corrosion in demanding hospital environments.

- **Front-loading door** equipped with a **safety interlock mechanism** prevents opening during operation, ensuring user protection.
- **Ergonomic design** allows easy loading/unloading of trays and racks.
- Integrated **touchscreen control panel** provides a clear, intuitive interface with step-by-step prompts for program selection and monitoring.

4.2 Chamber

The process chamber is designed for **efficient cleaning and optimized water/air circulation**.

- **Dimensions (H × W × D):** 625 × 620 × 570 mm
- **Volume:** 200 liters
- **Material:AISI 316L stainless steel**, chosen for its high corrosion resistance and suitability for medical device reprocessing.
- Smooth welded seams and rounded corners reduce dirt accumulation and facilitate chamber cleaning.

4.3 Integrated Systems

Bluvia Neo incorporates advanced systems for reliable and repeatable performance:

- **Water Softening & Regeneration Unit**: Prevents limescale deposits and enhances detergent performance. Automatic regeneration ensures continuous efficiency.
- **Deionized (DI) Water Rinse**: Provides residue-free rinsing, preventing mineral deposits and protecting delicate instruments.
- **HEPA-Filtered Forced-Air Drying**: Ensures rapid, contamination-free drying. Air passes through a **HEPA H14 filter**, maintaining sterility during the drying process.

4.4 Control System

The washer-disinfector is equipped with an advanced microprocessor-based control unit.

- **Touchscreen User Interface** with pre-set programs (Prime, Duo, Rapid, Rinsing, and Custom cycles).
- **Automatic cycle documentation**, with storage of process data such as temperature, time, and disinfection values.
- **User-level authentication** ensures secure operation, with roles for Operator and Administrator.
- **Built-in diagnostics** provide real-time error reporting, ensuring minimal downtime.

5. User Interface & Menu

The Bluvia Neo is operated through an **intuitive touchscreen interface**. The main menu provides access to validated programs, utility functions, logs, and system settings. Each option is designed for efficiency, safety, and compliance with hospital reprocessing protocols.

5.1 Cycle Programs

- **Prime Mode**
  The standard validated cycle for routine instrument reprocessing. Includes pre-rinse, enzymatic cleaning, neutralization, thermal disinfection ($A0 \geq 3000$), and HEPA-filtered drying. Recommended for general surgical instruments.
- **Duo Mode**
  Dual-stage cycle designed for **high-volume or heavily soiled loads**. Provides extended wash and disinfection phases to ensure reliable results.
- **Rapid Mode**
  Optimized cycle for **lightly soiled instruments** or quick turnaround needs. Shortened cleaning and drying times reduce total cycle duration while maintaining validated disinfection.
- **Custom Mode**
  Allows the operator to configure cycle elements according to need:
  - Rinse & Dry only
  - Disinfection & Drying only
  - DI Wash & Drying
  - Save Custom Cycle for repeated use

5.2 Utility Functions

- **Regeneration**
  Initiates the **automatic regeneration cycle** for the built-in water softener. Ensures consistent water quality and prevents scale deposits inside the chamber and on instruments.
- **Empty Cycle**
  Chamber-cleaning program to remove residual dirt, detergent, or biofilm. Recommended for routine hygienic maintenance.

5.3 Logs & Documentation

- **Cycle Log** – Records all cycle parameters (time, temperature, detergent dosing, A0 values).
- **Malfunction Log** – Stores error messages and fault history.
- **Batch Log** – Assigns Batch ID for traceability during audits.
- **Service Logs** – Maintains service history and maintenance tasks.

5.4 Monitoring & Control

- **Water Conductivity Check**
  Monitors DI water quality in real time, alerting if conductivity exceeds limits. Ensures proper rinsing performance.
- **Settings**
  Configurable options include:
  - Date & Time
  - Display brightness and volume
  - Language selection
  - User authentication & password management
  - Administrator permissions (cycle access, data export)
- **Maintenance Alerts**
  Provides automatic reminders for filter changes, salt refill, detergent replacement, or scheduled service.
- **Diagnosis & Info**
  Built-in diagnostic module with:
  - Error code explanations
  - Troubleshooting wizard for step-by-step problem solving
  - Certification reports for compliance audits
- **Cycle Progress Tracking**
  Real-time monitoring of active cycle with display of remaining time, current phase, and status indicators.

6. Preparing Loads

Correct preparation of instruments is essential for effective cleaning, disinfection, and drying. Improper loading may result in incomplete reprocessing, residual contamination, or damage to instruments.

6.1 Pre-treatment

- **Pre-rinse heavily soiled instruments** under running cold water immediately after use to prevent drying of blood and organic residues.
- Avoid using hot water at this stage, as it may cause protein coagulation and make cleaning more difficult.

6.2 Instrument Preparation

- **Hinged instruments**: Open fully to allow spray penetration.
- **Multi-part instruments**: Disassemble according to the manufacturer's instructions.
- **Delicate instruments**: Place in protective holders or silicone mats to prevent mechanical damage.

6.3 Tray & Rack Loading

- Place instruments in **DIN trays (50–100 mm height)** or other compatible baskets.
- Ensure hollow instruments (e.g., suction tubes, cannulas, endoscopic tools) are connected to **manifold adaptors** to guarantee internal lumen cleaning.
- Arrange instruments so they **do not overlap** or obstruct spray arms and airflow.
- Heavier items should be placed at the bottom, lighter and delicate ones at the top.

6.4 Load Capacity & Circulation

- Do not exceed the recommended load per cycle.
- Maintain sufficient **clearance on sides, back, and between trays** to ensure proper water and air circulation.
- Avoid stacking or tightly packing instruments, as this may compromise cleaning and drying performance.

7. Operating Instructions

The Bluvia Neo is designed for simple and safe operation. Only trained personnel should operate the device. Follow the steps below to ensure validated and reproducible results.

7.1 Starting the Device

1. Ensure the machine is connected to power and water supply.
2. Turn on the **main power switch**.
3. The **touchscreen control panel** will illuminate and display the home screen.

7.2 Loading Instruments

1. Prepare instruments as described in **Section 6 (Preparing Loads)**.
2. Open the front-loading door when the lock indicator is off.
3. Insert trays, racks, or adaptors carefully into the chamber.
4. Close the door until it clicks and the lock engages.

7.3 Selecting a Cycle

1. On the touchscreen, navigate to the **Main Menu**.
2. Choose the appropriate cycle based on load type and soil level:
    - **Prime** – Standard validated cycle
    - **Duo** – For large or heavily soiled loads
    - **Rapid** – For lightly soiled loads requiring fast turnaround
    - **Custom** – For user-defined settings
3. Confirm cycle selection.

7.4 Preparing Process Agents

- Verify that **detergent, neutralizer, rinse aid, and regeneration salt** containers are filled.
- Replace or refill if required.
- The system will display alerts if any agent is low or missing.

7.5 Running the Cycle

1. Press the **Start** button on the touchscreen.
2. The cycle begins automatically, with each phase displayed in real time.
3. Progress indicators show the current step (cleaning, disinfection, drying) and remaining time.

7.6 Completion of Cycle

1. At the end of the cycle, the system will display a completion message.
2. Wait for the **door unlock signal** before attempting to open.
3. Carefully remove trays and instruments. Instruments should be clean, dry, and ready for sterilization or direct use (as per protocol).

7.7 Aborting a Cycle

To stop a cycle in progress:

1. Press **Stop** on the touchscreen.
2. Confirm the action when prompted.
3. The chamber will drain automatically, and the system will return to standby mode.

Note: Instruments removed after an aborted cycle are **not considered disinfected** and must be reprocessed before use.

8. Logging & Documentation

The Bluvia Neo ensures complete **traceability and compliance** through automated logging of all reprocessing cycles. This feature supports hospital infection-control protocols and regulatory audit requirements.

8.1 Automatic Recording

For every completed or aborted cycle, the system automatically records:

- **Date and Time** of operation
- **Cycle Type** (Prime, Duo, Rapid, Custom, etc.)
- **Batch Identification Number (Batch ID)** assigned by the operator or system
- **Temperature Profile** and disinfection parameters
- **Validated A0 Value**, confirming thermal disinfection efficacy
- **Operator/User ID** (if authentication is enabled)

8.2 Log Storage & Export

- Logs are securely stored in the system memory.
- Records can be **exported via USB or SD card** in standardized formats for hospital records or external validation.
- Exported logs may include detailed reports (cycle graphs, error codes, maintenance data) for documentation.

8.3 Data Security & Integrity

- All logs are **digitally time-stamped** and protected against tampering.
- Data integrity is ensured through **encryption and secure storage protocols**.
- Only users with **administrator rights** can modify or delete stored records.

9. Function Checks & Maintenance

Regular maintenance ensures safe, efficient, and long-term operation of the **Bluvia Neo Washer-Disinfector**. The following schedule should be followed by trained personnel. Certain tasks must be performed only by qualified service engineers.

9.1 Daily Checks (Operator)

- **Salt Level:** Verify the regeneration salt container and refill if necessary.
- **Detergent/Neutralizer/Rinse Aid Levels:** Check containers and replace if low.
- **Empty Cycle:** Run a short empty cycle at the start of the day to flush the system and verify proper operation.
- **Door Seal & Surfaces:** Inspect the chamber door gasket and wipe chamber surfaces with a lint-free cloth if needed.

9.2 Weekly Checks (Operator)

- **Chamber Cleaning:** Wipe down internal chamber walls with a damp cloth and approved cleaning solution.
- **Spray Arms:** Remove and check spray arms for blockages; clean nozzles to maintain proper water circulation.
- **Water Inlet Filter:** Inspect and rinse the mesh filter to prevent debris buildup.

9.3 Monthly Checks (Operator/Technician)

- **HEPA Pre-Filter:** Replace or clean according to instructions to ensure effective air filtration during drying.
- **Seals & Gaskets:** Inspect door gasket and chemical tubing connections for wear or leakage. Replace if necessary.
- **Cycle Validation:** Review recent cycle logs for any irregularities or repeated error messages.

9.4 Annual Maintenance (Service Engineer)

- **Full System Validation:** Perform validation of cleaning and disinfection performance, including verification of A0 values, temperature sensors, and chemical dosing accuracy.
- **Calibration:** Check and calibrate sensors (temperature, conductivity, flow meters) as per service guidelines.
- **Mechanical Components:** Inspect pumps, valves, and circulation systems for wear or malfunction.
- **HEPA Filter Replacement:** Replace the main HEPA H14 filter to ensure sterile drying air.
- **Software & Firmware Updates:** Apply updates if available to maintain compliance and performance.

9.5 Maintenance Reminders

The Bluvia Neo provides **automatic maintenance alerts** on the touchscreen for:

- Low detergent, neutralizer, rinse aid, or salt
- Filter replacement due
- Scheduled service intervals

Important: Only authorized service personnel should perform electrical, mechanical, or software maintenance beyond daily/weekly operator checks.

10. Troubleshooting

Error Message	Cause	Action
Low Water Pressure	Insufficient inlet pressure	Check supply
Salt Empty	No regeneration salt	Refill salt
Detergent Empty	Container empty	Refill
Door Not Locked	Door not fully closed	Close properly
Conductivity High	DI water supply exhausted	Replace DI cartridge

11. Technical Specifications

Parameter	Specification
External Dimensions (H × W × D)	1950 × 740 × 765 mm
Chamber Volume	~ 200liters (useable volume)
Weight	~ 100 kg (without load)
Power Supply	230 V AC, 50 Hz, 3.5 kW
Water Supply	Cold water: 1–10 bar, ≤ 25 °C Hot water: 45–65 °C DI water: ≤ 15 µS/cm conductivity
Salt Tank Capacity	1.5 kg (medical grade regeneration salt)
Drying System	HEPA H14 filtered, forced-air drying; 40–120 °C adjustable
Cycle Temperatures	Cleaning: 45–60 °C Disinfection: 90–95 °C
Disinfection Value	Validated A0 ≥ 3000

12. Accessories & Consumables

The following accessories and consumables are recommended for safe and effective operation of the Bluvia Neo:

- Spray arm with Nozzle
- Single-level manifold rack (DIN standard compatible)
- Mesh trays (50–100 mm) for flexible instrument loading
- **Enzyme-alkaline detergent** (validated for protein removal)
- **Neutralizer** (acid-based, for alkalinity balance)
- **Rinse aid** (low-foaming, medical grade)
- Medical grade regeneration salt for water softening unit
- HEPA H14 filter (for sterile drying air)
- Water inlet mesh filter (removable and cleanable)
- Outlet hose
- Exhaust hose

- Exhaust mount
- Power cable and user manual

13. Pause Time & Decommissioning

- **Short Pauses (≤ 24 hours):** Leave the chamber door slightly open to allow ventilation and prevent odor or condensation.
- **Extended Pauses (> 2 weeks):**
  - Drain residual water from tanks and chamber.
  - Disconnect power supply and water inlets.
  - Store in a clean, dry, dust-free environment.
- **Permanent Decommissioning / Disposal:**
  - Remove and properly dispose of process agents (detergents, neutralizers, rinse aid, salt).
  - Disconnect all utilities (power, water, drain).
  - Return unit to an authorized **medical equipment recycler** in accordance with local environmental regulations.

14. Regulatory & Compliance

The Bluvia Neo Washer-Disinfector is manufactured to meet international medical device standards and regulatory frameworks:

- **ISO 15883:** Washer-disinfector standards (general requirements, surgical instruments, and test methods).
- **ISO 13485:** Quality Management System for medical device manufacturing.
- **Electrical Safety & EMC:** Designed in compliance with IEC 60601 and IEC 61010 safety standards.
- **CDSCO Submission Reference:** *Bluvia Neo W/D – Medford Technologies Pvt Ltd*.

15. Glossary

- **A0 Value** – A standard measure of the thermal disinfection effect, representing equivalent microbial inactivation achieved at 80 °C.
- **HEPA Filter** – *High Efficiency Particulate Air* filter capable of capturing ≥ 99.995% of airborne particles (class H14).
- **DI Water** – *Deionized water*, with conductivity ≤ 15 µS/cm, used for final rinsing to prevent mineral deposits.

- **DIN Tray** – Standardized instrument tray (per German Institute for Standardization), ensuring compatibility across washer racks.
- **User Manual** – Official instruction document for safe operation, maintenance, and compliance of the Bluvia Neo Washer-Disinfector.